Nexio CRM: TZ va joriy loyiha solishtirmasi

Manba: Nexio_CRM_Full_Plan_with_8_Screens.pdf (v2.0, 12.10.2026). Loyiha: nexio_crm_v2 (main, 43a9031). Tekshiruv sanasi: 09.11.2026. Baholash kod (React src/, Laravel backend/) bo'yicha statik tahlil asosida; ishga tushirib sinalmagan.

Jami talablar: 54 | **Bor: 25** | **Qisman: 21** | **Yo'q: 8**

1. 8 ta asosiy ekran (PDF 2-9 betlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
01 Dashboard	**Bor**	Dashboard.tsx + GET /dashboard: KPI, savdo grafigi (Recharts), voronka, vazifalar, AI faollik, so'nggi lid/buyurtmalar; qo'shimcha MyDay sahifasi	KPI kartalaridan filtrli o'tish hali yo'q (davr filtri URL'da saqlanadi - YANGI)
02 Login / Welcome	**Bor**	Login, Register, parolni tiklash, email tasdiqlash, taklif (invite), workspace almashtirish (switch-tenant); YANGI: MFA/TOTP (RFC 6238 testlangan; sozlamalar > Xavfsizlik; resurs kodlari, qayta ishlatishdan va brute-force'dan himoya), /choose-workspace (bir nechta kompaniya bo'lsa login keyin), /onboarding (haqiqiy ma'lumotga asoslangan 4 qadamli checklist, ro'yxatdan keyin)	Alohida /mfa marshruti yo'q (kod kiritish login sahifasida ikkinchi bosqich sifatida)
03 Leads & Pipeline	**Bor**	Lidlar ro'yxati, yaratish, konvertatsiya, diskvalifikatsiya; Deals kanban (drag&drop), /deals/:id/move, CSV eksport; YANGI: CSV import wizard (ko'rib chiqish + validatsiya + formula-inyeksiya himoyasi), /app/leads/:id va /app/deals/:id detail sahifalari; kanban dnd-kit'ga o'tkazildi (sichqoncha, sensorli ekran va klaviatura bilan ko'chirish)	-
04 Clients / Companies	**Bor**	Companies, CompanyDetail (tablar), Contacts, faoliyat (activities); YANGI: /app/contacts/:id detail; custom fields (Sozlamalar > Dop. maydonlar, turlar: matn/son/sana/ro'yxat/ha-yo'q, majburiylik, tenant bo'yicha izolyatsiya); kontakt<->kompaniya ko'p-ko'p bog'lanish rollar bilan; izohlar (notes) va fayllar tablari (10 MB, ruxsat etilgan turlar, tarif kvotasi, shaxsiy saqlash)	-
05 Projects & Orders	**Bor**	Projects, ProjectDetail, milestones, Tasks, Orders, buyurtma tasdiqlash, buyurtmadan loyiha yaratish; YANGI: Gantt (loyihalar vaqt shkalasi), /app/calendar (vazifa/loyiha/bitim sanalari), /app/orders/:id detail; fayllar/izohlar (loyiha, buyurtma, vazifa); vazifa vaqti hisobi (time entries); vazifa bog'liqliklari (sikl himoyasi, bloklangan vazifani boshlab/yopib bo'lmaydi)	-

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
06 AI Control Center	Qisman	AI agentlar (CRUD, status), kampaniya start/stop, qo'ng'iroqlar ro'yxati/tafsilot, approvals, suppression list, AI analitika, Groq LLM, telephony plagini; YANGI: /app/ai-knowledge (bilim bazasi hujjatlari, agentga yoki hammaga, LLM javobiga ulangan, tekshiruv paneli), /app/ai-policies (siyosat muharriri; 'AI ekanini aytish' o'chirib bo'lmaydi, chegirma/urinish/tinch soat validatsiyasi), /app/ai-usage (kun/agent bo'yicha sarf, tarif limitlari), /app/ai-campaigns/:id (voronka, natijalar, qo'ng'iroqlar)	Faqat real ovozli streaming (ASR/TTS + SIP) yo'q - qo'ng'iroqlar sandbox/simulate; bu tashqi provayder shartnomasini talab qiladi
07 Billing & Subscriptions	Bor	Tariflar, tarif almashtirish, bekor/qayta tiklash, invoice to'lash, Click webhook (imzo, dublikatdan himoya), avto-uzaytirish; YANGI: Payme Merchant API (JSON-RPC: Check/Create/Perform/Cancel/CheckTransaction/GetStatement, Basic auth, tiyin summa, 12 soat muddat, bir martalik hisob-kitob), to'lov usulini tanlash (Click/Payme), /app/billing/transactions (filtr, sahifalash, jami), qo'shimcha modullar ro'yxati; 7 ta avtomatik test	Payme real kabinet (sandbox) bilan hali sinalmagan: merchant ID va kalitlarni ulab, Payme sinov to'plamidan o'tkazish kerak
08 Platform Super Admin	Bor	Alohida PlatformShell: overview, tenants, plans, subscriptions, usage, Nexio buyurtmalar, support, AI-ops, security, settings, marketplace, audit API	/platform/health sahifasi va health-check monitoringi yo'q

2. Qo'shimcha marshrutlar (PDF 4-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Auth marshrutlari	Bor	/login, /register, /reset-password, /verify-email, /invite/:token, YANGI: /choose-workspace, /onboarding	Alohida /mfa marshruti (funksional jihatdan login ichida)
CRM detail sahifalari	Bor	/app/companies/:id, YANGI: /app/leads/:id, /app/contacts/:id, /app/deals/:id	-
Operatsiyalar	Qisman	YANGI: calendar; orders, YANGI: orders/:id, projects/:id, tasks, communications, helpdesk (support), documents	/app/files
Moliya	Qisman	/app/finance (umumiy), /app/billing, /app/catalog (products)	Alohida /app/invoices, /app/payments, /app/expenses, /app/usage
Hisobot va avtomatlashtirish	Qisman	/app/analytics, /app/automations (dry-run bilan), /app/integrations	/app/reports, /app/api-keys, /app/webhooks
AI marshrutlari	Bor	/app/ai-agents, /ai-agents/:id, /ai-calls, /ai-analytics, YANGI: /ai-campaigns/:id, /ai-knowledge, /ai-usage, /ai-policies	-
Jamoa va sozlamalar	Qisman	/app/team (rollar/ruxsatlar), /app/settings (logo, brend rangi, til, vaqt zonasi)	Alohida /app/roles va /app/settings/security
Nexio xizmatlari / portal	Bor	/app/nexio-services, mijoz portali /portal	-

3. Dizayn-tizim (PDF 3-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Rang tokenlari	Qisman	#2563EB / #1D4ED8 va boshqa ranglar CSS o'zgaruvchilarida (--color-*)	--nx-* nomlash standarti yo'q; qo'shimcha tungi tema mavjud (TZ'da faqat light)
Inter shrifti	Qisman	font-family: Inter, system-ui ...	Self-hosted WOFF2 Inter ulanmagan (faqat tizimdagi bo'lsa ishlaydi)
Lucide ikonkalar, Recharts	Bor	lucide-react, recharts	-
Holatlar: skeleton, empty, error, toast	Bor	UI komponentlar: core, table, toast, overlay, form	prefers-reduced-motion qo'llab-quvvatlashi topilmadi
RU / UZ / EN interfeys	Bor	O'z i18n provayderi (en/ru/uz), standart: uz	i18next ishlatilmagan (TZ talabi); barcha sahifalar to'liq tarjima qilinganini tekshirish kerak

4. Frontend steki (PDF 5-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
React + TS + Vite + Router	Bor	React 18, TypeScript, Vite 6, react-router 6, lazy loading	-
TanStack Query, Tailwind, Zod	Bor	@tanstack/react-query, Tailwind 4, zod	-
Zustand, shadcn/Radix, RHF, TanStack Table, dnd-kit	Qisman	dnd-kit (kanban) o'rnatildi; o'zimizning UI komponentlar	Zustand, shadcn/Radix, React Hook Form, TanStack Table o'rnatilmagan
Testlar (Vitest, RTL, Playwright)	Qisman	Vitest: 3 ta unit test (api, format, timezones)	React Testing Library va Playwright e2e yo'q
Kod tuzilmasi (apps/web, entities, features, packages/ui)	Yo'q	src/pages, src/components, src/lib	Monorepo, alohida apps/admin, packages/ui, OpenAPI client yo'q

5. Backend - Laravel (PDF 6-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
PHP 8.3 + Laravel + PostgreSQL	Bor	Laravel 13, PHP ^8.3, DB_CONNECTION=pgsql (SqlDialect orqali)	-
Redis, Horizon, S3	Yo'q	Queue/cache = database	Redis, Horizon, S3 obyekt saqlash sozlanmagan
Sanctum / OAuth2	Yo'q	O'z sessiya mexanizmi (sessions jadvali, ApiContext middleware)	Sanctum, OAuth2 third-party yo'q
Modul tuzilmasi (Policy, FormRequest, Event, Job)	Qisman	Domen bo'yicha controllerlar + Rbac, H, Ctx yordamchilari; Plugin tizimi (EventBus, Lifecycle)	Policies, FormRequests, Repository/Event/Job qatlamlari yo'q; controllerlar katta
Identity / Tenancy	Bor	Users, tenants, memberships, invitations, switch-tenant, 14 tenant roli, MFA	-

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
CRM / Sales	Qisman	Organizations, contacts, leads, pipelines, deals, activities, products, orders	Quotes, price_lists, tags yo'q
Projects	Qisman	Projects, milestones, tasks, attachments, time entries, task dependencies	Project members, comments yo'q
Finance / Subscription	Qisman	Invoices, payments, plans, subscriptions, usage_records, billing_events, prorating, past_due/grace	Expenses, credit notes, refund, currency_rates jadvali yo'q (CBU kurs servisi bor)
Integration	Qisman	Integrations toggle, webhook_events, Telegram bot, plugin marketplace	API tokenlar (scoped), outbound webhooks (HMAC, retry, DLQ) yo'q
Audit	Bor	audit_logs, /settings/audit, plugin audit	-
OpenAPI spetsifikatsiya	Yo'q	-	OpenAPI 3.1 schema, Postman/Bruno kolleksiya yo'q

6. Multi-tenancy, rollar va xavfsizlik (PDF 9, 14-bo'limlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
tenant_id izolyatsiyasi	Bor	Barcha so'rovlarda tenant_id filtri	PostgreSQL RLS yo'q
Rollar va ruxsatlar matritsasi	Bor	Rbac.php: tenant_owner, admin, sales_manager, finance_manager va b.	Scope (own/team/all) darajasini tekshirish kerak
Tenant isolation testlari	Qisman	PluginTenantIsolationTest + 9 ta boshqa backend test	CRM uchun to'liq A/B tenant izolyatsiya testlari yo'q
MFA, rate limit, CSP/HSTS	Qisman	YANGI: MFA (TOTP) + MFA urinishlariga limit (5 ta / 5 daqiqa); resend-verification throttle	Umumiy login rate limit, CSP/HSTS sozlamalari topilmadi

7. Nexio AI Agents (PDF 11-bo'lim)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Rozilik va DNC	Bor	consent_calls, suppression_list, do_not_contact, compliance gate	-
Qo'ng'iroq state machine	Qisman	queued, compliance_checked, dialing, connected, ai_disclosed, conversation ...	Provider event id bilan idempotent callbacklar real provayderda tekshirilmagan
Voice Sales Agent (real qo'ng'iroq)	Yo'q	Telephony plagini (HTTP provider + sandbox), simulate call	Real SIP/telefoniya + ASR/LLM/TTS streaming gateway yo'q
Buyurtmani AI tasdiqlashi + approval	Qisman	ai_approvals, finalize call, ai_tool_calls	Narx/chegirma limitlari to'liq tekshiruvini tasdiqlash kerak
Xarajat limitlari	Qisman	Agentda budget_limit	Kampaniya/kun/oy bo'yicha hard stop tekshirilmagan

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Analytics/Support/Ops agentlari	Qisman	AI analitika sahifasi, AI assistant plagini (Groq)	Alohida Support va Ops agentlari yo'q

8. Integratsiyalar, DevOps va hujjatlar (PDF 13, 14, 18-bo'limlar)

Talab (TZ)	Holat	Loyihada bor	Yetishmaydi / qilish kerak
Telegram	Bor	Bot, webhook, polling, foydalanuvchini bog'lash	-
To'lov provayderlari	Bor	Click (imzo bilan), Payme (YANGI), sandbox; invoice hisob-kitobi umumiy InvoiceSettlement xizmatida (idempotent)	Payme real kabinet bilan sinalmagan
Web-forma SDK / widget	Yo'q	-	Saytdan lid olish widgeti yo'q
Docker Compose, CI/CD	Yo'q	deploy.sh skripti	docker-compose, Dockerfile, GitHub Actions yo'q
Monitoring (Sentry, health, alerts), backup	Yo'q	-	Sentry, alerting, backup/restore runbook yo'q
Hujjatlar	Qisman	PLUGIN_ARCHITECTURE.md, NEXIO_INVENTORY*.md, .env.example, seederlar	ARCHITECTURE, SECURITY, TENANCY, OPERATIONS, AI_POLICIES .md, RU/UZ/EN qo'llanmalar yo'q
TZ'dan tashqari qo'shimchalar	Bor	Plugin marketplace + developer portal, Inventory (ombor), Delivery (Uzum, Yandex), Helpdesk, Documents (taklif generatsiyasi), Automations	-

Asosiy xulosa: MVP uchun eng muhim yetishmayotganlar

1. (Bajarildi) MFA, /onboarding, /choose-workspace.
2. To'liq tenant izolyatsiya testlari (CRM uchun A/B tenant) - TZ bo'yicha release gate.
3. Quotes (tijorat takliflari), price lists, tags, project members.
4. Umumiy /app/files sahifasi, S3 saqlash (hozir lokal shaxsiy disk).
5. Scoped API tokenlar, outbound webhooks (HMAC), OpenAPI spetsifikatsiya.
6. Docker Compose, CI/CD, monitoring (Sentry/health), backup; Redis/queue worker.
7. AI: real telefoniya + ovozli gateway (hozircha sandbox/simulate); Payme real kabinetda sinash.